



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 10

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UNIT TEST - 3, NOVEMBER - 2024

SCIENCE

Time Allowed : 1.30 Hours]

[Max. Marks : 50

PART - I

I. Choose the best answer.

9x1= 9

- The frequency which is audible to the human ear is
a) 50 KHZ b) 20 KHZ c) 15000 KHZ d) 10,000 KHZ
- The sound waves are reflected from an Obstacle into the same medium from which they were incident. Which of the following changes?
(a) Speed (b) Frequency (c) Wave Length (d) None of above
- $H_2 + Cl_2 \rightarrow 2 HCl$ is a
a) Decomposition reaction b) Combination reaction
c) Single Displacement reaction d) Double Displacement reaction
- The P^H of a solution is 3, Its $[OH^-]$ concentration is
a) $1 \times 10^{-3}M$ (b) 3 M (c) $1 \times 10^{-11}M$ (d) 11M
- The 'Use and disuse' theory was proposed by -----
a) Charles Darwin b) Ernst Haeckel
c) Jean Baptiste Lamarch d) Gregor Mendel
- The best way of direct dating fossils of recent origin is by
(a) Radio - Carbon method (b) Uranium Lead method
(c) Potassium - Argon method (d) Both (a) and (c)
- r DNA is a
(a) Vector DNA (b) Circular DNA
(c) Recombinant of Vector DNA and desired DNA (d) Satellite DNA
- The Miracle rice which saved millions of lives and celebrated 1st 50th birthday is
a) IR8 b) IR24 c) Atomita 2 d) Ponni
- We can cut the DNA with the help of
(a) Scissors (b) Restriction Endonucleases
(c) Knife (d) RNA ase

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PART - II

II. Answer any four of the following questions. (Q.No.15 is compulsory)

4x2=8

10. Why does Sound Travel Faster on a Rainy day than on a dry day?

11. Fill in the blanks:

a) Chemical Volcano is an example for ----- type of reaction.

b) The normal P^H of Human blood is -----

12. Why is Archaeopteryx Considered to be the Fossil Bird?

13. Name the types of Stem Cells.

14. What is Metastasis? *Mention two casts in which there no Doppler effect*

15. What will be the Frequency of sound having 0.20m as its Wavelength? When it travels with a speed of 331ms^{-1}

PART - III

III. Answer any three of the following. (Q.No.20 is compulsory)

3x4=12

16. a) What is an Echo?

b) State two Conditions necessary for hearing an Echo.

17. Differentiate between Reversible reaction and Irreversible reactions.

18. Write the Importance of Ethnobotany.

19. The Degenerated wing of a Kiwi is an Acquired Character. Why is it an Acquired Character.

20. What is the P^H of 1.0×10^{-5} Molar Solution of KOH?

PART - IV

IV. Answer all the questions.

3x7=21

21. a) (i) What do you understand by the term "Ultrasonic Vibration"?

(ii) State the uses of Ultrasonic Vibrations.

(iii) Name three animals which can hear Ultrasonic Vibrations.

(OR)

b) What are the Factors that affect the speed of sound in gases? Explain.

22. a) Explain the Factors Influencing the rate of a reaction.

(OR)

b) Explain the types of double displacement reactions with examples.

23. a) With a neat labelled diagram explain the techniques involved in Gene Cloning.

(OR)

b) Discuss the importance of Biotechnology in the field of Medicine.